

THE RATIOS SYSTEM: ANALYSIS OF ROI (RETURN ON INVESTMENT)

Chapter 6



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A first distinction:

- **ROA** = Net Income / Total Assets
- «**ROI**» = Operating Income / Operating Assets (operating profitability)

The main decisions about the operating profitability ratio are:

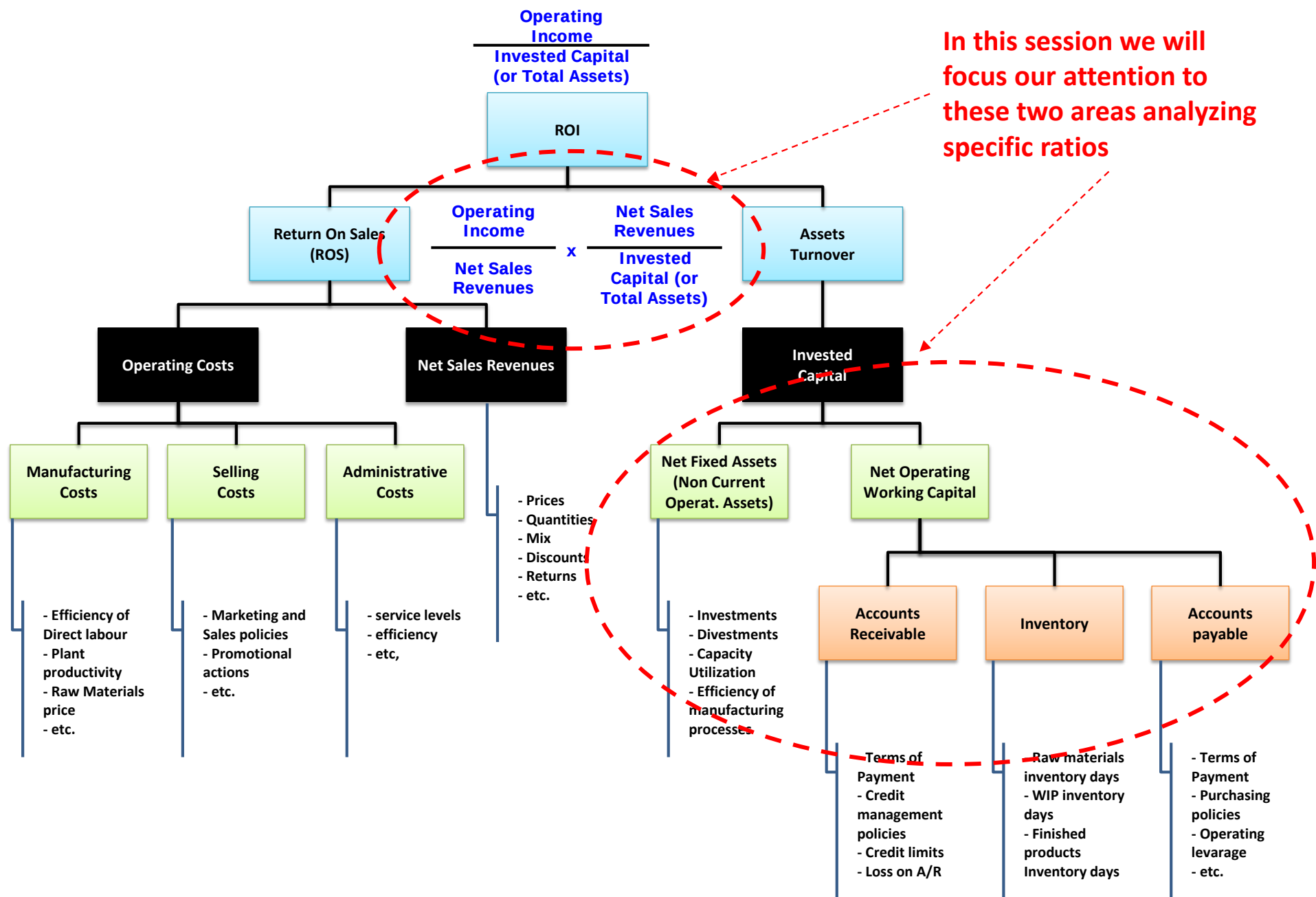
- **Operating Income**: core vs. non core; before vs. after taxes
- **Operating Assets**: core vs. non core; gross vs. net (net of operating liabilities)

The operating profitability ratio can be called in different ways:

- **ROI**, Return on Investment
- **ROIC**, Return on Invested Capital
- **ROCE**, Return on Capital Employed
- **RONA**, Return on Net Assets

The name is not important. It is important to clarify what you put both at the numerator and at the denominator !!!

Anyway, the most frequent name for the operating profitability ratio are: ROIC and RONA



$$\text{R.O.I.} = \frac{\text{Operating Income}}{\text{Total Invested Capital}} = \left(\frac{\text{Core Operating Income}}{\text{Core Operating Invested Capital}} \times \frac{\text{Core Operating Invested Capital}}{\text{Total Invested Capital}} \right) + \left(\frac{\text{Non Core Operating Income}}{\text{Non Core Operating Invested Capital}} \times \frac{\text{Non Core Operating Invested Capital}}{\text{Total Invested Capital}} \right)$$

2= Non Core Operating Profitability Ratio

Core Operating Income		Core Operating Income		Net Sales Revenues
	=		x	
Core Operating Invested Capital		Net Sales Revenues		Core Operating Invested Capital

ROI decomposition, based on Total **Net** Invested Capital (net of Operating Liabilities, always subtracted from the Core Operating Assets):

$$\text{R.O.I.} = \frac{\text{Operating Income}}{\text{Total Net Invested Capital}} = \left(\frac{\text{Core Operating Income}}{\text{Core Operating Net Invested Capital}} \times \frac{\text{Core Operating Net Invested Capital}}{\text{Total Net Invested Capital}} \right) + \left(\frac{\text{Non Core Operating Income}}{\text{Non Core Net (?) Operating Invested Capital}} \times \frac{\text{Non Core Net (?) Operating Invested Capital}}{\text{Total Net Invested Capital}} \right)$$

1 2

1 = Core (net) Operating Profitability Ratio (from sales), the typical RONA (Return on Net Operating Assets)

2= Non Core Operating Profitability Ratio

$$\frac{\text{Core Operating Income}}{\text{Core Operating Net Invested Capital}} = \frac{\text{Core Operating Income}}{\text{Net Sales Revenues}} \times \frac{\text{Net Sales Revenues}}{\text{Core Operating Net Invested Capital}}$$

$$\frac{\text{Net Sales Revenues}}{\text{(gross) Core Operating Invested Capital}} \times \frac{\text{(gross) Core Operating Invested Capital}}{\text{Core Operating Net Invested Capital}}$$

gross

net

$$\frac{\text{(gross) Core Operating Invested Capital}}{\text{Core Operating Net Invested Capital}} = \frac{1}{1 - \frac{\text{Operating Liabilities}}{\text{(Gross) Core Operating Invested Capital}}}$$

Higher than 1 when there are Operating Liabilities

A first ratio that measures the «Operating Liability Leverage»



The data

Net Sales Revenues	16,00
Operating Costs	-13,60
Core Operating Income	2,40
Non (Core) Operating Income	0,15
Operating Income	2,55

5,00	Current Operating Assets	Operating Liabilities	4,00
7,00	Non Current Operating Assets	Financial Liabilities	4,00
3,00	Non Core Operating Assets (or Non Operating Assets)	Net Equity	7,00
15,00			15,00

Gross R.O.I. = $\frac{\text{Operating Income}}{\text{Invested Capital}} = \left(\frac{\text{Core Operating Income}}{\text{Core Operating Invested Capital}} \times \frac{\text{Core Operating Invested Capital}}{\text{Total Invested Capital}} \right) + \left(\frac{\text{Non Core Operating Income}}{\text{Non Core Operating Invested Capital}} \times \frac{\text{Non Core Operating Invested Capital}}{\text{Total Invested Capital}} \right)$

$$\frac{2,55}{15,00} = \left(\frac{2,40}{12,00} \times \frac{12,00}{15,00} \right) + \left(\frac{0,15}{3,00} \times \frac{3,00}{15,00} \right)$$

$$17,00\% = (20,00\% \times 80,00\%) + (5,00\% \times 20,00\%)$$

Net R.O.I. = $\frac{\text{Operating Income}}{\text{Net Invested Capital}} = \left(\frac{\text{Core Operating Income}}{\text{Core Operating Net Invested Capital}} \times \frac{\text{Core Operating Net Invested Capital}}{\text{Net Invested Capital}} \right) + \left(\frac{\text{Non Core Operating Income}}{\text{Non Core Operating Net Invested Capital}} \times \frac{\text{Non Core Operating Net Invested Capital}}{\text{Net Invested Capital}} \right)$

$$\frac{2,55}{11,00} = \left(\frac{2,40}{8,00} \times \frac{8,00}{11,00} \right) + \left(\frac{0,15}{3,00} \times \frac{3,00}{11,00} \right)$$

$$23,18\% = (30,00\% \times 72,73\%) + (5,00\% \times 27,27\%)$$



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Core Operating Income	2,40
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Operating Income	2,55

$$\frac{\text{Core Operating Income}}{\text{Core Operating Net Invested Capital}} = \frac{\text{Core Operating Income}}{\text{Net Sales Revenues}} \times \frac{\text{Net Sales Revenues}}{\text{Core Operating Net Invested Capital}}$$

$$\frac{2,40}{8,00} = \frac{2,40}{16,00} \times \frac{16,00}{8}$$

$$30,0\% = 15,0\% \times 2,00$$

Impact of the «Operating Liability Leverage»

But...

5,00	Current Operating Assets	Operating Liabilities	4,00
7,00	Non Current Operating Assets	Financial Liabilities	4,00
3,00	Non Core Operating Assets (or Non Operating Assets)	Net Equity	7,00
15,00			15,00

Also the decomposition of Core Net ROI is different

$$\frac{\text{Core Operating Income}}{\text{Core Operating Net Invested Capital}} = \frac{\text{Core Operating Income}}{\text{Net Sales Revenues}} \times \frac{\text{Net Sales Revenues}}{\text{Core Operating Gross Invested Capital}} \times \frac{\text{Core Operating Gross Invested Capital}}{\text{Core Operating Net Invested Capital}}$$

$$\frac{2,40}{8,00} = \frac{2,40}{16,00} \times \frac{16,00}{12,00} \times \frac{12,00}{8,00}$$

$$30,0\% = 15,0\% \times 1,33 \times 1,50$$

Net R.O.I. = $\frac{\text{Operating Income}}{\text{Net Invested Capital}} = \left(\frac{\text{Core Operating Income}}{\text{Core Operating Net Invested Capital}} \times \frac{\text{Core Operating Net Invested Capital}}{\text{Net Invested Capital}} \right) + \left(\frac{\text{Non Core Operating Income}}{\text{Non Core Operating Net Invested Capital}} \times \frac{\text{Non Core Operating Net Invested Capital}}{\text{Net Invested Capital}} \right)$

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$$23,18\% = (30,00\% \times 72,73\%) + (5,00\% \times 27,27\%)$$



$$\text{ROI} = \text{RoS} \times \text{Assets Turnover}$$

Analysis of the first ratio (**Return on Sales**) is based on the analysis of the reformulated Income Statement.

In order to analyze the **Assets Turnover Ratio**, we have to focus our attention to the 2 main components of Operating Assets:

1. Current Operating Assets
2. Non Current Operating Assets (or Net Fixed Assets)

Current Operating Assets are usually analyzed together with Operating Liabilities (mainly Accounts Payable) in order to understand the size and variability of **NOWC** in comparison with the size and trend of Sales Revenues and other Operating Income components. So, the focus of this analysis is usually on 3 items: A/R, Inventory and A/P.

Non Current Operating Assets (Tangible and Intangible Assets) are usually analyzed in order to understand:

- ☐ Capacity utilization (under-capacity vs. full-capacity)
- ☐ Correlation between the sales and manufacturing volumes and the size of investments (critical for capital intensive companies)



Non Current Operating Assets (Tangible and Intangible Assets) are usually analyzed in order to understand:

- ☐ Capacity utilization
- ☐ Correlation between the sales and manufacturing volumes and the size of investments (critical for capital intensive companies)

From the financial statements it is impossible to get data about the *capacity utilization* of PPE (extent or level to which the productive capacity of a plant, firm, or country is being used in generation of goods and services; expressed usually as a percentage, it is computed by comparing the total capacity with the portion being utilized).

The Fixed Assets Turnover can be a proxy of the *capacity utilization* metric

$$\text{Fixed Assets Turnover} = \frac{\text{Net Revenues}}{\text{Fixed (tangible) Assets}}$$

The denominator in this ratio can include *intangible assets* as well as tangible ones. This adjustment is intuitively appealing for firms that report substantial amount of intangibles, such as knowledge-based companies.

$$\text{Long-term Productive Assets Turnover} = \frac{\text{Net Revenues}}{\text{Tangible and Intangible Assets}}$$

Use caution when analyzing productive assets ratio tends. This measure exhibits more instability than short-term asset utilization ratios (NOWC ratios). Unlike current assets, fixed assets are acquired infrequently but in large increments. Therefore, this rate of turnover may extremely differ from one period to the next.

Remember the effects on Net Fixed Assets of: depreciation policies and impairment!!!

$$\Delta \text{Tangible(Intangible) Assets} = + \text{Acquisitions} - \text{Disposal} - \text{Depreciation} - \text{Impairment}$$



What is the **correlation between the sales and manufacturing volumes and the size of investments?**

Let's analyze the change in the amount of Net Fixed Assets because of investments/divestments, i.e. the **Capital Expenditures (CAPEX)**; CAPEX are the amount of acquisitions of tangible and intangible assets (net of disposals).

In some capital-intensive industries, every year companies make investments in order to support to business. We can observe the size of these net investments with two ratios:

$$\text{CAPEX-to-Sales (or Capital Intensity ratio)} = \frac{\text{CAPEX}}{\text{Net Revenues}}$$

Correlation of Net Investments with the operating size of business (sales revenues)

$$\text{Productive Capacity Replacement} = \frac{\text{CAPEX}}{\text{Depreciation and Amortization}}$$

Correlation of Net Investments with the consumption of existing Fixed Assets (depreciation)

Analyst would like to identify to different component of (net) investments:

1. **Productive Capacity Maintenance Investments**
2. **Productive Capacity Development (growth) Investments**

The two above ratios can help the analyst in identifying the investment rate (maintenance + development) of a company in comparison with industry average and best-in-class companies



So, you have to look at two circumstances:

1. If **CAPEX-to-Revenues** is aligned to the industry average or it is higher/lower than the industry average; ex. if it is higher that means that the company is investing more then the industry average and, consequently, there are probably some development investments.
2. If **CAPEX-to-Depreciation and Amortization** is close to 1 that means that investments are just replacing the consumption of existing long-term operating assets (maintenance investments); if, on the contrary, CAPEX-to-Depreciation and Amortization is **significantly higher than 1** that means that investments exceed the current consumption of existing long-term operating assets: the surplus to 1 are development investments and the company is expected to show a growth in the future.

We will come back to the analysis of CAPEX talking about the Cash Flow Statement

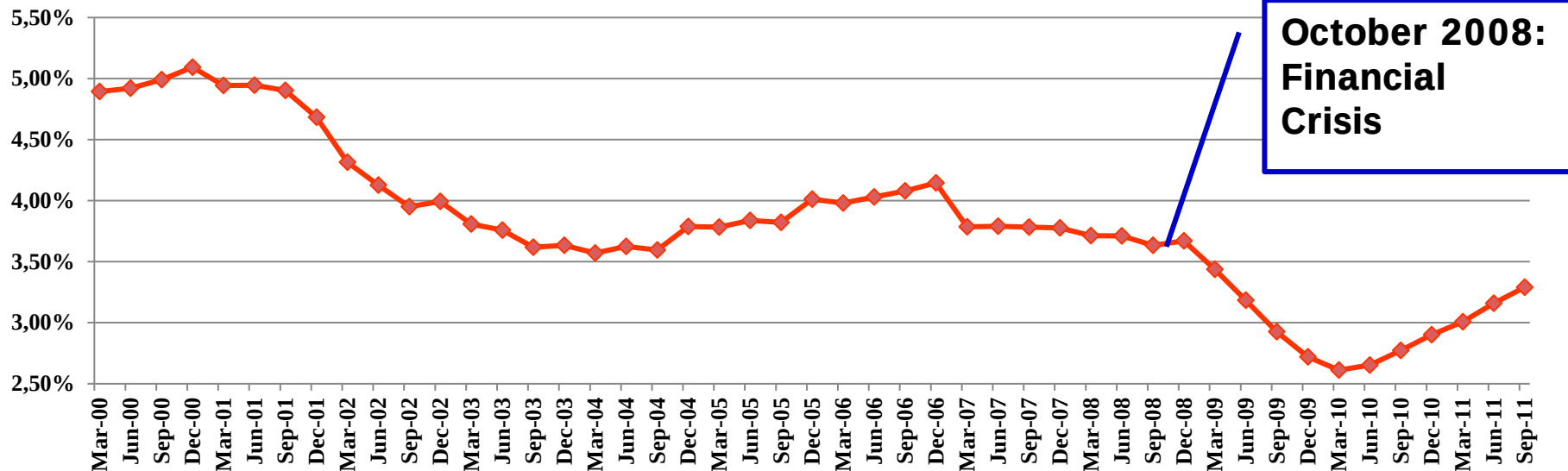
LET'S SEE NOW SOME EMPIRICAL DATA ➔





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Capital Expenditures / Revenue (%)

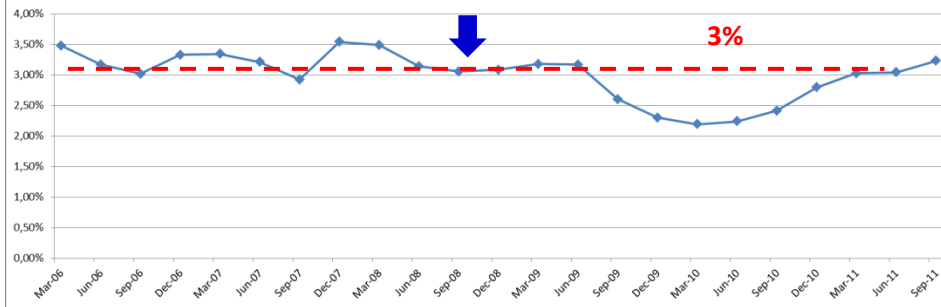


Source: **Cash Flow Trends and Their Fundamental Drivers: Comprehensive Industry Review (Quarter 3, 2011)**. All companies with a current market capitalization of \$50 million or more are included, resulting in a total sample of 2,893 companies



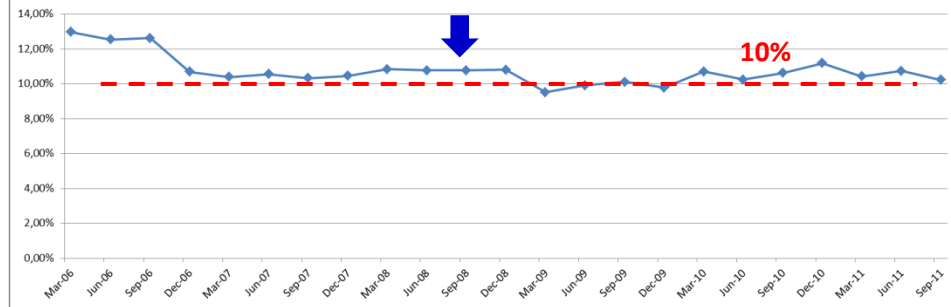
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GRADUATE SCHOOL

Capital Expenditures / Revenue (%)



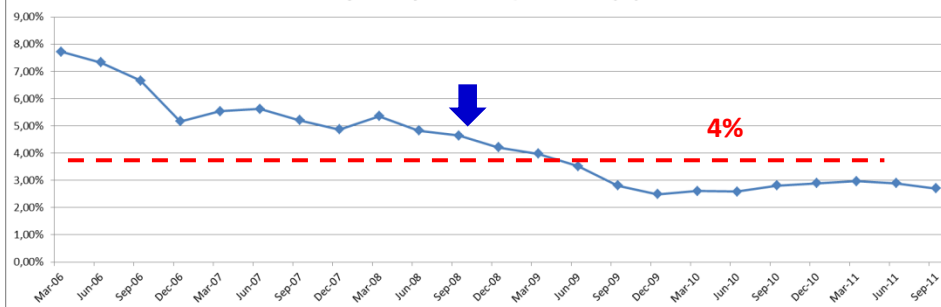
CICS Industry: 23, Automobiles and Trucks

Capital Expenditures / Revenue (%)



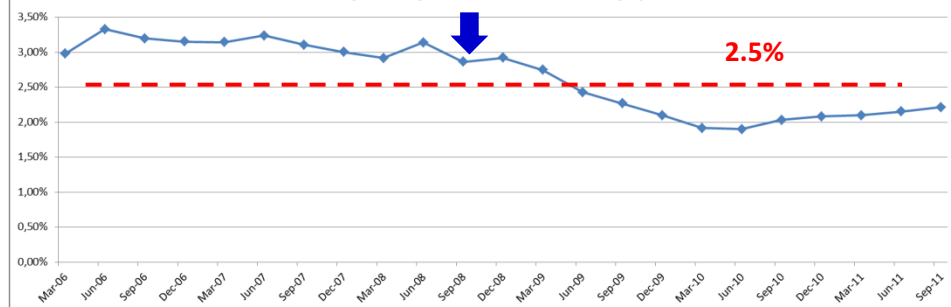
CICS Industry: 32, Communication

Capital Expenditures / Revenue (%)



CICS Industry: 13, Pharmaceutical Products

Capital Expenditures / Revenue (%)



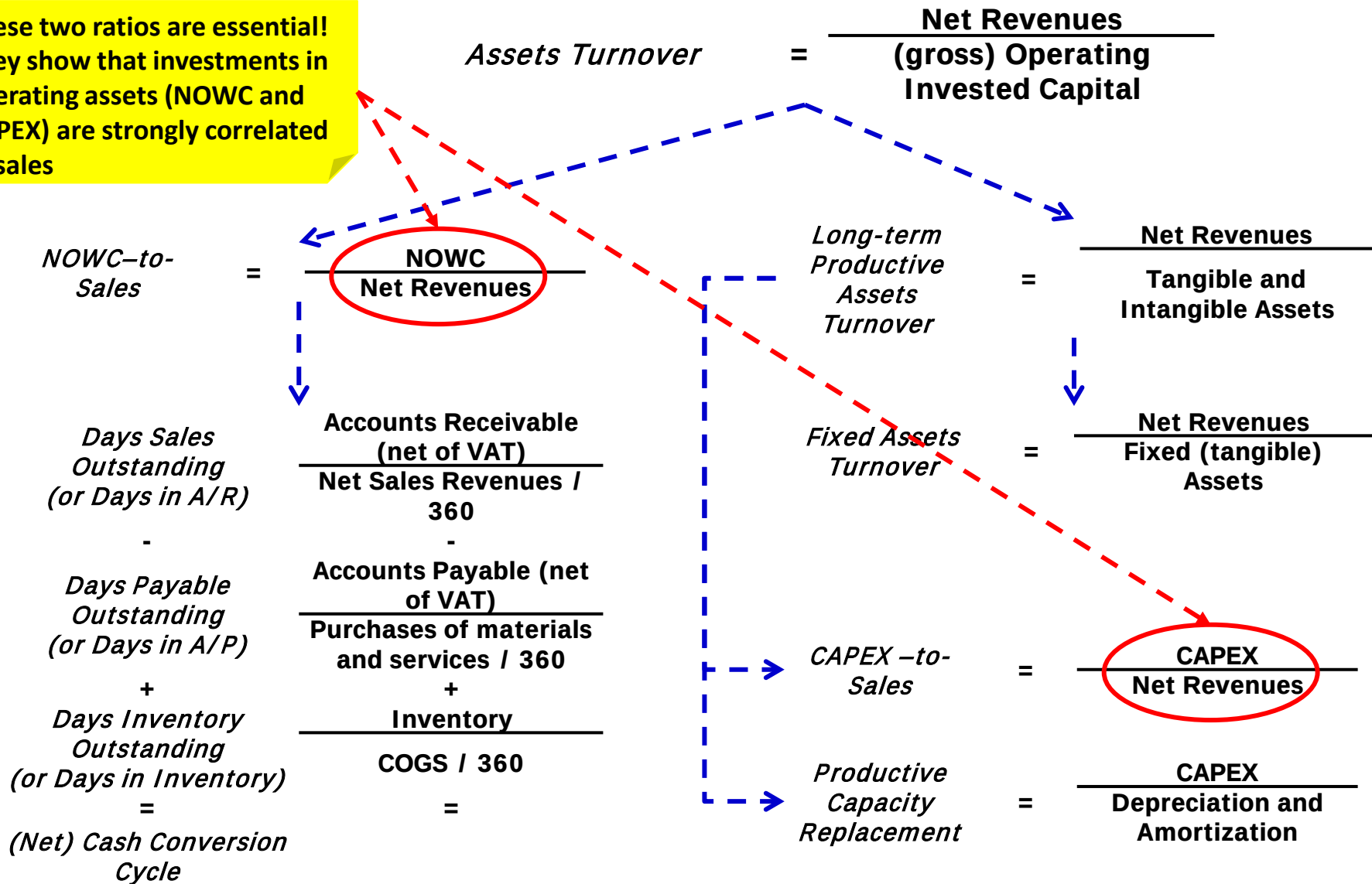
CICS Industry: 43, Retail

A couple of comments:

- The Automobiles industry showed a strong reaction to the financial crisis (they immediately cut CAPEX and they came back to a 3% average only in 2011)
- The Communication industry (including Telco companies) did not cut investments after the financial crisis: because of the continuous investment in new mobile and data technologies, if you do not want to exit from the market...



These two ratios are essential!
They show that investments in operating assets (NOWC and CAPEX) are strongly correlated to sales



	A	B	C		A	B	C
Revenues	100,0	100,0	100,0	➔	120,0	120,0	120,0
COGS (60%), of which:	60,0	60,0	60,0		72,0	72,0	72,0
Purchases (50% of COGS)	30,0	30,0	30,0		36,0	36,0	36,0
Gross Profit	40,0	40,0	40,0		48,0	48,0	48,0
Days in A/R	90,0	45,0	30,0		90,0	45,0	30,0
Days in Inventory	60,0	➔ 30,0	➔ 20,0		60,0	30,0	20,0
Days in A/P (manufacturing)	-120,0	-120,0	-180,0		-120,0	-120,0	-180,0
Cash Conversion Cycle	30,0	-45,0	-130,0		30,0	-45,0	-130,0
Accounts Receivable	25,0	12,5	8,3		30,0	15,0	10,0
Inventory	10,0	5,0	3,3		12,0	6,0	4,0
Accounts Payable	-10,0	-10,0	-15,0		-12,0	-12,0	-18,0
NOWC	25,0	7,5	-3,3		30,0	9,0	-4,0
NOWC / Revenues	25,0%	7,5%	-3,3%		25,0%	7,5%	-3,3%
					5,0	1,5	-0,7
					20,0	20,0	20,0
					25,0%	7,5%	-3,3%

All things being equal (Cash Conversion Cycle, COGS / Revenues and Purchases / COGS), a change in Revenues does **NOT** change the value of the ratio:
NOWC / Revenues !!!

The ratio **NOWC / Revenues** has a relevant predictive value in order to understand how the size of NOWC changes accordingly to a change in Net Revenues

How can we interpret the Cash Conversion Cycle (CCC)?

The traditional interpretation is: a positive number of days, ex. 30 days, means that the company needs to get financing to cover 30 days of investments in A/R and Inventory, net of the financial support provided by the suppliers. But 1 day of A/R is different, in Euros, from 1 day in A/P e and 1 day in inventory....So, the CCC ratio is misleading!!!

What can we say if this ratio is negative? Does a negative CCC ratios imply that NOWC is negative? A negative CCC doesn't necessarily imply that the NOWC is negative! Only in extreme scenarios (CCC extremely negative) the NOWC can be negative

Many analysts prefer the ratio NOWC / Net Revenues. Why? Because there is a strong correlation between these two amounts, so this ratios has a significant predictive value, in both alternatives: NOWC / Revenues; Δ NOWC / Δ Revenues

Is it true that a growth in sales revenues implies a growth in NOWC? Yes, a growth in revenues implies, all things being equal, a growth in NOWC (investment that needs more financing) EXCEPT for extreme scenarios where the CCC ratio and NOWC are extremely negative (like in CASE 3 above).



Coherently with the calculation of **Net Adjusted Assets (elimination of Goodwill and other intangibles with indefinite useful life for which no amortization is calculated)**, it is possible to calculate the ratio as:

$$\frac{\text{Core Operating Income}}{\text{Net Adj. Operating Assets}} = \frac{\text{Core Operating Income}}{\text{NOWC} + \text{Tangible Assets} + \text{Specific Intangibles}}$$

Specific intangibles are **recurring intangibles with a definite useful life** (so, amortization is required); ex. Development costs, software, intellectual property rights etc.

By eliminating Goodwill and Trademarks we do not measure the return on the full amount of investments, including those originated by acquisitions, BUT:

- **Goodwill and trademarks may refer to acquisitions made many years ago...and they are “frozen” (no amortization) and changes are possible only for impairments**
- **The Net Adj. Operating Assets, at the opposite, show every year some changes: investments net of depreciation and amortization, often strongly correlated with the amount of Net Revenues (NOWC/Net Revenues; CAPEX/Net Revenues).**



Let's suppose that you are analyzing a single investments (like the opening of a new store). There is a big initial investment followed by small expenses for renewal and replacement due to obsolescence. Look at this ratio:

$$\text{Return on Gross Operating Assets} = \frac{\text{Core Operating Income}}{\text{Gross Operating Assets (before Accumulated Depreciation)}}$$

Gross Operating Assets: The total historical or original cost of assets used in operations (such as gross property, plant, and equipment plus working capital assets), ignoring accumulated depreciation.

Why Use Gross Operating Assets?

- ***Eliminates Age Bias:*** Using gross values stops older assets with high accumulated depreciation from artificially inflating the return ratio compared to newer assets.
- ***Consistent Asset Base:*** It keeps the asset denominator stable at historical cost, making it easier to compare different divisions or periods without depreciation distortions.

Remember: for “old” investments (stores, factories) the Net Book Value of long-term assets can be very small or null (only in managerial accounting, for internal analysis, you can estimate the replacement cost of these assets and the related depreciation).

If we reformulate the Income Statement in order to divide operating costs in some relevant categories, it could be possible to apply the same criteria to the analysis and decomposition of ROI.

Remember: mathematical decompositions must be useful and not (too much) complex”.

Let's an example: the so called **GMROI (gross margin return on inventory)**, a ratio that is normally used in the retail industry, considering this reformulation of Income Statement:

Net Sales Revenues

- Cost of Sales (or COGS): purchases of merchandise net of discounts, commercial cooperation services and ancillary and logistics costs +/- change in merchandise inventory

= Gross Margin (or Gross Profit)

- Payroll Expenses

- Other purchases

- Depreciation and amortization

- Provisions

- Other Expenses

= Operating Income



In € millions	FY 2025	FY 2024	Δ %
Revenue	32,142	31,669	+1.5%
Gross margin	7,374	7,303	+1.0%
% revenue	22.9%	23.1%	-0.1 pt
Payroll expenses	-4,323	-4,294	+0.7%
External expenses	-2,099	-2,155	-2.6%
% revenue	-20.0%	-20.4%	+0.4 pt
EBITDA	1,025	883	+16.1%
% revenue	3.2%	2.8%	+0.4 pt

□ Gross Margin Percentage is:

$$\text{Gross Margin Percentage} = \frac{\text{Gross Margin}}{\text{Net Revenues}}$$

$$\text{Auchan (2025)} = \frac{7,374}{32,142} = 22,9\%$$

$$\text{Auchan (2024)} = \frac{7,303}{31,669} = 23,1\%$$

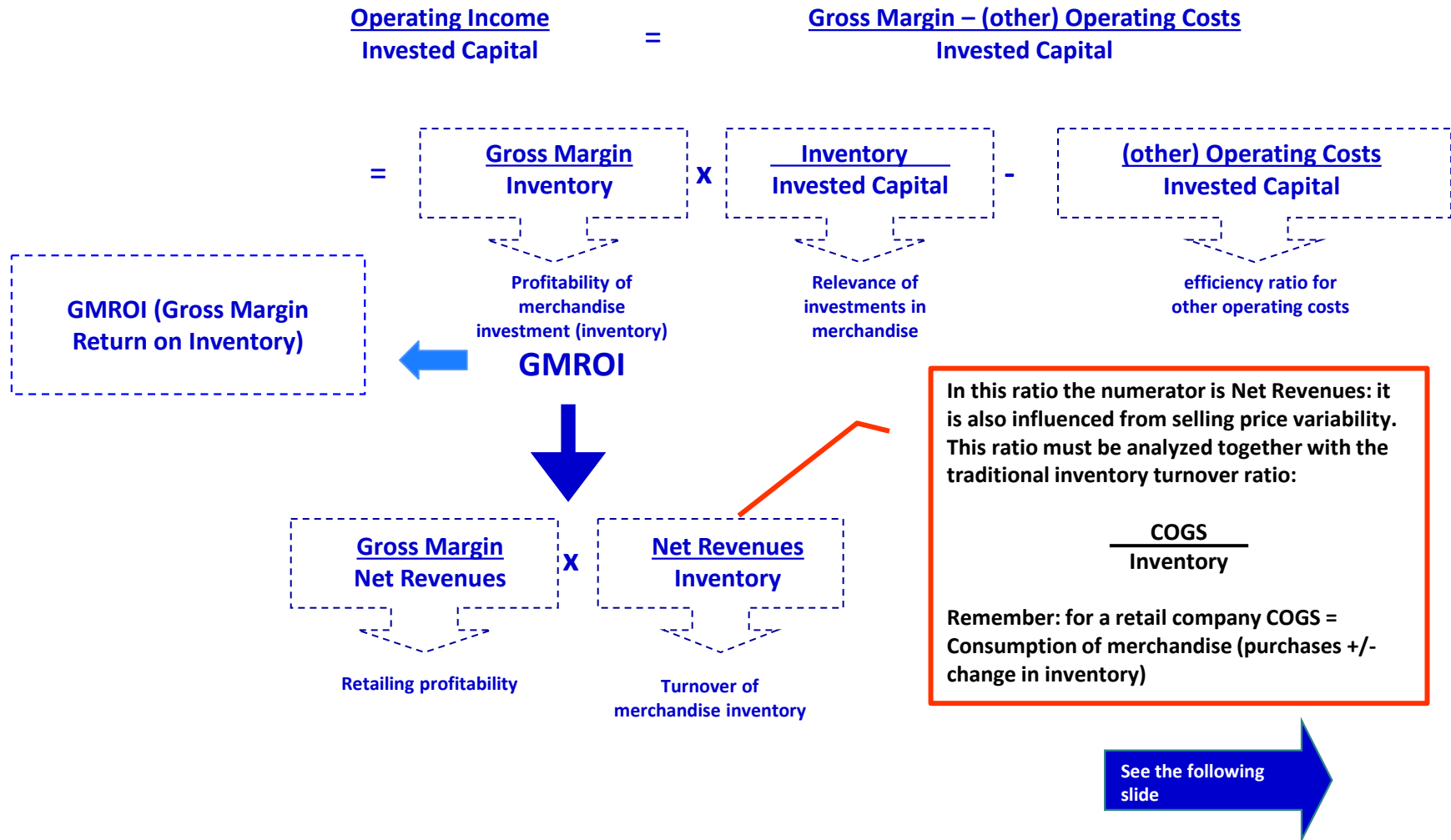
□ Remember:

$$\text{gross margin} = \frac{\text{markup}}{1 + \text{markup}}$$

$$\text{markup} = \frac{\text{gross margin}}{1 - \text{gross margin}}$$

2025	22,9% / (1-22,9%) =	29,7%
2024	23,1% / (1-23,1%) =	30,0%





Gross Margin Return on Inventory (GMROI) is a "turn and earn" metric that measures inventory performance based on both margin and inventory turnover. In essence, GMROI answers the question, "For every dollar carried in inventory, how much is earned in gross profit?" GMROI can be calculated at the organization level and, if the proper data is collected at the item level, all the way down to an individual item.

$$\text{GMROI} = \frac{\text{Gross Profit}}{\text{Sales}} \times \frac{\text{Sales}}{\text{Avg Inventory at cost}}$$

The *earn* term, \$ Gross Profit / \$ Sales, is simply the gross profit margin % (GPM%).

The *turn* term, \$ Sales / \$ Average Inventory at cost, is not the same as inventory turnover (ITO), which is the \$ Cost of Sales or Cost of Goods Sold divided by \$ Average Inventory at cost.

The GMROI formula can be simplified to address this difference and use Inventory Turnover (ITO) with the following formula:

$$\text{GMROI} = \text{GPM\%} \times \frac{\text{ITO}}{1 - \text{GPM\%}}$$

Sales/Inventory
 = COGS/Inventory X Sales/COGS →
 = COGS/Inventory : COGS/Sales →
 = COGS/Inventory : (Sales/Sales – Gross Margin/Sales) →
 = COGS/Inventory : (1 – Gross Margin/Sales)

This can be restated as:

$$\text{GMROI} = \frac{\text{GPM \%}}{\text{CGS \%}} \times \text{ITO}$$

where CGS % = Cost of Goods Sold % or 1 - GPM %.

For example:

A company has Sales of \$10,000 and Gross Profit of \$1,000. The GPM % = 10%.

The Cost of Sales is \$9,000 and the CGS % = 90%.

If the average inventory is \$900, then ITO = 10.

This company earns 1.11 in Gross Profit for every dollar carried in inventory (\$1,000 Gross Profit / \$900 average inventory).

Using the restated formula, $\text{GMROI} = \frac{\text{GPM \%}}{\text{CGS \%}} \times \text{ITO},$

$$\text{GMROI} = \frac{10\%}{90\%} \times 10 \text{ or } 0,11 \times 10 = 1,11$$

